PHASE 2: COMPLETE STUDY GUIDE

Numbers + Permutation & Combination + Probability | Target: 1.5 Hours | The Heavy Hitters

TOPIC 1: NUMBERS (30 minutes)

1A. TYPES OF NUMBERS (2 min scan)

Natural: 1,2,3... | Whole: 0,1,2,3... | Integers: ...,-2,-1,0,1,2,...

Even: divisible by 2 | Odd: not divisible by 2

Prime: only divisible by 1 and itself (2,3,5,7,11,13,17,19,23,29,31,37,41,43,47...)

Rational: can be written as a/b (b!=0) | Irrational: non-terminating, non-recurring decimals

Co-prime: Two numbers whose HCF = 1. (e.g., 49 and 64 are co-prime)

Key fact: *Primes > 3 are ALWAYS of form 6n +/- 1. To check if N is prime, test divisibility by primes up to sqrt(N).*

Worked Example (from your PDF Q7):

Is 113 prime? sqrt(113) ~ 10.6. Check primes up to 10: 2, 3, 5, 7.

113/2 = no | 113/3 = no | 113/5 = no | 113/7 = no. So **113 IS prime**.

1B. SUMMATION FORMULAS (3 min - MEMORIZE)

Sum of first n natural numbers = n(n+1)/2

Sum of first n perfect squares = n(n+1)(2n+1)/6

Sum of first n perfect cubes = $[n(n+1)/2]^2$

Sum of first N ODD naturals = N^2

Sum of first N EVEN naturals = N(N+1)

Worked Example (from your PDF Q3):

Sum of first 35 natural numbers = 35 x 36 / 2 = **630**

Worked Example (from your PDF Q4):

Sum of odd numbers up to 40? There are 20 odd numbers up to 40.

Sum = 20^2 = **400**

Worked Example (from your PDF Q15 - Handshakes):

30 people shake hands with everyone. Total handshakes?

Each shakes with 29 others. But don't double-count! = 29+28+27+...+1 = Sum of first 29 = 29x30/2 = **435**

Shortcut: *Handshakes = nC2 = n(n-1)/2. This connects to Combinations (Phase 2 later)!*

1C. SUM & PRODUCT TRICKS (2 min read)

Max product when sum is fixed: Split into EQUAL parts

Min sum when product is fixed (integers): Take extreme values

Worked Example (from your PDF Q5):

Sum of two numbers = 28. Maximum product?

Equal split: 14 x 14 = **196**

Worked Example (from your PDF Q6):

Product of 2 integers = 36. Minimum possible sum?

Take -1 and -36. Sum = -1 + (-36) = **-37**

Worked Example (from your PDF Q16):

Product of 4 integers = 72. Maximum possible NEGATIVE sum?

Take -9, 8, -1, 1. Product = 72. Sum = -9+8-1+1 = **-1**

1D. DIVISIBILITY RULES (5 min - CRITICAL)

These are tested EVERY exam. Memorize the table:

By	Rule	Example
2	Last digit is 0,2,4,6,8	138 -> 8 is even -> Yes
3	Sum of digits divisible by 3	23781 -> 2+3+7+8+1=21 -> Yes
4	Last TWO digits divisible by 4	145896 -> 96/4=24 -> Yes
5	Last digit is 0 or 5	895 -> ends in 5 -> Yes
6	Divisible by BOTH 2 AND 3	Must be even + digit sum div by 3
8	Last THREE digits divisible by 8	135128 -> 128/8=16 -> Yes
9	Sum of digits divisible by 9	92754 -> 9+2+7+5+4=27 -> Yes
11	Diff(odd place sum, even place sum) = 0 or 11k	65967 -> (6+9+7)-(5+6)=11 -> Yes
12	Divisible by BOTH 3 AND 4	Check digit sum + last 2 digits
25	Last TWO digits divisible by 25	Ends in 00, 25, 50, or 75

Worked Example (from your PDF Q8):

654#62 is divisible by 3. How many values of #?

Digit sum = 6+5+4+#+6+2 = 23+#. For div by 3: 23+# = 24,27,30. So # = 1, 4, 7. Answer: **3**

Worked Example (from your PDF Q19 - HARD, divisibility by 11):

27942*@ is divisible by 11. * and @ are different digits. How many numbers?

Odd places: 2+9+2+@ = 13+@. Even places: 7+4+* = 11+*.

Diff = (13+@)-(11+*) = 2+@-* = 0 or 11 or -11

Case 1: @-* = -2 => (0,2),(1,3),(2,4),(3,5),(4,6),(5,7),(6,8),(7,9) = 8 pairs

Case 2: @-* = 9 => only (9,0) = 1 pair. Total = **9**

1E. FACTORS OF A NUMBER (3 min read)

If N = $a^p \times b^q \times c^r$ (prime factorization), then:

Number of factors = (p+1)(q+1)(r+1)

Sum of factors = $[(a^{p+1}-1)/(a-1)] \times [(b^{q+1}-1)/(b-1)] \times [(c^{r+1}-1)/(c-1)]$

Product of factors = $N^{(\text{number of factors} / 2)}$

Worked Example (from your PDF Q14):

Find number of factors of 150.

150 = $2^1 \times 3^1 \times 5^2$. Factors = (1+1)(1+1)(2+1) = 2x2x3 = **12**

Worked Example (from your PDF Q20):

N = $2^5 \times 3^2 \times 5^1$. Find sum and product of factors.

Number of factors = (5+1)(2+1)(1+1) = 36

Sum = $[(2^6-1)/(2-1)] \times [(3^3-1)/(3-1)] \times [(5^2-1)/(5-1)]$ = 63 x 13 x 6 = **4914**

Product = $N^{36/2} = N^{18} = \mathbf{2^{90} \times 3^{36} \times 5^{18}}$

1F. REMAINDER TRICK (2 min read)

For products: Find individual remainders, multiply them, then take remainder again

Worked Example (from your PDF Q17):

Remainder when 65347 x 74789 x 57743 is divided by 4?

Only last 2 digits matter for div by 4. 47%4=3, 89%4=1, 43%4=3.

Product of remainders = 3x1x3 = 9. Remainder of 9/4 = **1**

NUMBERS: PRACTICE QUESTIONS

N1. Sum of first 25 even natural numbers?

- a) 625
- b) 650
- c) 600
- d) 675

N2. Greatest 4-digit number divisible by 12?

- a) 9996
- b) 9984
- c) 9988
- d) 9990

N3. How many factors does 720 have?

- a) 24
- b) 28
- c) 30
- d) 32

N4. 87649#64 is divisible by 4. How many values of #?

- a) 5
- b) 8
- c) 10
- d) 3

N5. Sum of two numbers is 36. Maximum product?

- a) 320
- b) 324
- c) 360
- d) 330

N6. Which pair is co-prime? (a) 195,215 (b) 77,143 (c) 49,64 (d) 31,93

- a) 195,215
- b) 77,143
- c) 49,64
- d) 31,93

NUMBERS: SOLUTIONS

N1. Answer: b) 650 | Sum of first N even = N(N+1) = 25 x 26 = 650.

N2. Answer: a) 9996 | 9999/12 = 833 remainder 3. So 9999-3 = 9996.

N3. Answer: c) 30 | 720 = 2^4 x 3^2 x 5^1. Factors = (4+1)(2+1)(1+1) = 30.

N4. Answer: c) 10 | Div by 4 depends on last 2 digits only: 64/4=16. Already divisible! So # can be 0-9 = 10 values.

N5. Answer: b) 324 | Equal split: 18 x 18 = 324.

N6. Answer: c) 49,64 | 49=7^2, 64=2^6. No common factor. HCF=1. Co-prime!

TOPIC 2: PERMUTATION & COMBINATION (30 minutes)

2A. FUNDAMENTAL COUNTING (2 min read)

AND (both jobs) => MULTIPLY: m x n ways

OR (either job) => ADD: m + n ways

Worked Example (from your PDF Ex1 & Ex2):

10 boys, 8 girls. Select a boy AND a girl = 10 x 8 = **80**
10 boys, 8 girls. Select a boy OR a girl = 10 + 8 = **18**
Memory trick: *AND = multiply, OR = add. Simple as that.*

2B. PERMUTATION (Arrangement - ORDER MATTERS) (5 min)

nPr = n! / (n-r)!

n objects all at a time = n!

Factorials: 1!=1, 2!=2, 3!=6, 4!=24, 5!=120, 6!=720, 7!=5040, 0!=1

With REPEATED letters:

n! / (p! x q! x r!) where p,q,r are repetitions

Worked Example (from your PDF - MISSISSIPPI):

MISSISSIPPI: 11 letters. M=1, I=4, S=4, P=2.
Arrangements = 11! / (4! x 4! x 2!) = 39916800 / (24 x 24 x 2) = **34,650**

Worked Example (from your PDF Q3 - Numbers div by 4):

5-digit numbers from 1,2,3,4,5,6 (no repetition) divisible by 4?
For div by 4, last 2 digits must form number div by 4:
Possible endings: 12,16,24,32,36,52,56,64 = 8 cases
Remaining 3 digits from 4 left: 4x3x2 = 24 each. Total = 8 x 24 = **192**

Worked Example (from your PDF Q4):

4-letter words from EQUATION, must include N?
N is fixed. Choose 3 more from 7 remaining: 7C3 = 35. Arrange all 4: 4! = 24.
Total = 35 x 24 = **840**

2C. CIRCULAR PERMUTATION (2 min read)

n objects in a circle = (n-1)! ways

Necklace/Bracelet (no flip distinction) = (n-1)! / 2

Worked Example (from your PDF Q7):

10 different beads in a necklace?
Necklace => (10-1)! / 2 = **9!/2** = 181440

Worked Example (from your PDF - 8 students in circle):

8 students around a circle = (8-1)! = **7!** = 5040

2D. DICTIONARY RANK (3 min - IMPORTANT)

To find the rank of a word among all arrangements in dictionary order:

Count all words that come BEFORE it alphabetically.

Worked Example (from your PDF Q8 - Rank of MOTHER):

Letters of MOTHER: E, H, M, O, R, T (alphabetical)
Words starting with E_ _ _ _ _ = 5! = 120
Words starting with H_ _ _ _ _ = 5! = 120

Words starting with ME_ _ _ _ = 4! = 24
Words starting with MH_ _ _ _ = 4! = 24
Words starting with MOE_ _ _ = 3! = 6
Words starting with MOH_ _ _ = 3! = 6
Words starting with MOR_ _ _ = 3! = 6
MOTEHR = 1, MOTERH = 1, MOTHER = 1
Rank = 120+120+24+24+6+6+6+1+1+1 = **309**

2E. COMBINATION (Selection - ORDER DOESN'T MATTER) (3 min)

nCr = n! / [r! x (n-r)!]

nCr = nC(n-r) | nC0 = nCn = 1 | nC1 = n

Select 1 or more from n items = 2^n - 1

Quick Values to MEMORIZE:

nCr	1	2	3	4	5
n=4	4	6	4	1	-
n=5	5	10	10	5	1
n=6	6	15	20	15	6
n=7	7	21	35	35	21
n=10	10	45	120	210	252

Key Applications:

Handshakes among n people = nC2 = n(n-1)/2

Triangles from n points (no 3 collinear) = nC3

Parallelograms from M & N parallel lines = MC2 x NC2

Worked Example (from your PDF Q13 - Triangles from 2 lines):

10 points on line 1, 11 on line 2. How many triangles?
Pick 2 from line1, 1 from line2: 10C2 x 11C1 = 45 x 11 = 495
Pick 1 from line1, 2 from line2: 10C1 x 11C2 = 10 x 55 = 550
Total = 495 + 550 = **1045**

2F. ARRANGEMENT WITH CONSTRAINTS (2 min)

Worked Example (from your PDF Q18 - Boys & girls alternate):

4 boys and 4 girls in a row, alternating?
Case 1: BGBGBGBG => 4! x 4! = 576
Case 2: GBGBGBGB => 4! x 4! = 576
Total = **1152**

Worked Example (from your PDF Q15 - Vowels never together):

TRIANGLE: 8 letters, 3 vowels (I,A,E). Words where all vowels NOT together?
Total arrangements = 8! = 40320
Vowels together: treat 3 vowels as 1 block => 6 units x 6! x 3! = 6 x 720 x 6 = 4320
Vowels NOT together = 8! - 6!x6 = 40320 - 4320 = **36000**
Strategy: *'Never together' = Total - 'Always together'. This indirect method is usually faster.*

PERMUTATION & COMBINATION: PRACTICE QUESTIONS

PC1. 3 candidates for Classical, 5 for Math, 4 for Science. Ways to award scholarships?

a) 12
b) 60
c) 30
d) 120

PC2. How many February calendars needed for all future possibilities?

a) 7
b) 12
c) 14
d) 28

PC3. In a classroom, total handshakes = 66. How many students?

a) 10
b) 11
c) 12
d) 13

PC4. Arrangements of letters in LEADER?

a) 720
b) 360
c) 180
d) 240

PC5. 6 tasks, 6 persons. Task 1 not for person 1 or 2. Task 2 must be person 3 or 4. Ways?

a) 144
b) 180
c) 192
d) 360

PC6. 10 beads in a necklace. Arrangements?

a) 9!
b) 9!/2
c) 10!
d) 10!/2

PERMUTATION & COMBINATION: SOLUTIONS

PC1. Answer: b) 60 | AND case: 3 x 5 x 4 = 60.

PC2. Answer: c) 14 | Feb can have 28 or 29 days (2 options) x start on any of 7 days = 14.

PC3. Answer: c) 12 | nC2 = 66 => n(n-1)/2 = 66 => n(n-1) = 132 => n = 12.

PC4. Answer: b) 360 | 6 letters, E repeated twice. 6!/2! = 720/2 = 360.

PC5. Answer: a) 144 | Task 2: 2 ways. Task 1: 3 ways (not P1, P2, or whoever got Task 2). Rest: 4! = 24. Total = 2x3x24 = 144.

PC6. Answer: b) 9!/2 | Necklace = circular + no flip distinction = (n-1)!/2 = 9!/2.

TOPIC 3: PROBABILITY (30 minutes)

3A. THE CORE FORMULA (2 min read)

P(Event) = Favorable outcomes / Total outcomes

0 <= P(A) <= 1 | P(A) + P(not A) = 1

P = 1 means certain | P = 0 means impossible

3B. COINS, DICE & CARDS BASICS (3 min - MEMORIZE)

Experiment	Total Outcomes	Key Facts
1 Coin	2 (H, T)	P(H) = P(T) = 1/2
n Coins	2^n	3 coins = 8 outcomes
1 Die	6 (1-6)	P(any face) = 1/6
2 Dice	36	Sum=7 has 6 ways (most common!)
52 Cards	52	4 suits x 13 each
		4 Kings, 4 Queens, 4 Jacks, 4 Aces
		26 Red (Heart+Diamond), 26 Black (Spade+Club)
		13 each: A,2,3,4,5,6,7,8,9,10,J,Q,K

Worked Example (from your PDF Q1):

Dice thrown. P(getting 6) = 1/6. Simple!

Worked Example (from your PDF Q2):

3 coins tossed. P(exactly 1 or 2 heads)?

Total = 8. Exactly 1H: HHT,HTH,THH = nope, those are 2H. Exactly 1H: HTT,THT,TTH = 3. Exactly 2H: HHT,HTH,THH = 3.

Favorable = 3+3 = 6. P = 6/8 = **3/4**

Worked Example (from your PDF Q4):

Two dice. P(sum = 7)?

Pairs: (1,6)(2,5)(3,4)(4,3)(5,2)(6,1) = 6. P = 6/36 = **1/6**

Exam favorite: *Sum of 7 is the MOST likely sum with 2 dice. Memorize P = 1/6.*

3C. ADDITION & MULTIPLICATION RULES (3 min)

P(A or B) = P(A) + P(B) - P(A and B)

If mutually exclusive: P(A or B) = P(A) + P(B) [since P(A and B) = 0]

If independent: P(A and B) = P(A) x P(B)

THE MOST POWERFUL TRICK:

P(at least one) = 1 - P(NONE)

This is easier 90% of the time. Instead of adding all the 'at least 1' cases, just find 'none' and subtract from 1.

Worked Example (from your PDF Q10):

3 students. Chances of solving: 1/2, 1/3, 1/4. P(problem solved)?

P(none solves) = (1 - 1/2)(1 - 1/3)(1 - 1/4) = 1/2 x 2/3 x 3/4 = 6/24 = 1/4

P(at least one) = 1 - 1/4 = **3/4**

Worked Example (from your PDF Q13 - Odds based):

Odds 11:5 against man A surviving. Odds 5:3 against man B surviving.

P(A survives) = 5/(11+5) = 5/16. P(B survives) = 3/(5+3) = 3/8.

P(at least one survives) = 1 - P(neither) = 1 - (11/16)(5/8) = 1 - 55/128 = **73/128**

Odds a:b AGAINST means P(event) = b/(a+b). *Don't mix up!*

3D. CARD-BASED PROBABILITY (3 min)

Worked Example (from your PDF Q3):

1 card drawn from 52. P(King) = 4/52 = 1/13
P(Queen of Hearts) = 1/52
P(King or Queen) = 8/52 = 2/13

Worked Example (from your PDF Q5):

3 cards drawn. P(King, Queen, Jack)?
Favorable = 4C1 x 4C1 x 4C1 = 64. Total = 52C3 = 22100.
P = 64/22100 = **16/5525**

Worked Example (from your PDF Q9 - 4 cards from 4 suits):

4 cards dropped. P(all from different suits)?
Favorable = 13C1 x 13C1 x 13C1 x 13C1 = 13^4 = 28561
Total = 52C4 = 270725. P = 28561/270725 = **2197/20825**

3E. SPECIAL PROBABILITY TYPES (3 min)

Leap Year Problems:

Leap year = 366 days = 52 weeks + 2 extra days.

The 2 extra days can be: (Sun,Mon), (Mon,Tue), (Tue,Wed),..., (Sat,Sun) = 7 equally likely pairs.

P(53 Sundays in leap year) = 2/7 (Sunday appears in 2 of 7 pairs)

Worked Example (from your PDF Q6):

P(53 Mondays in leap year) = 2/7 (Monday appears in (Sun,Mon) and (Mon,Tue))

Minimum Tosses Problem:

Worked Example (from your PDF Q16):

Min coin tosses for P(at least 1 head) >= 0.9?
P(at least 1H) = 1 - $(1/2)^n$ >= 0.9 => $(1/2)^n$ <= 0.1
n=3: 1/8=0.125 > 0.1 (fails). n=4: 1/16=0.0625 <= 0.1 (works!). Answer: **4**

Conditional / Bag Problems:

Worked Example (from your PDF Q12 - Two bags):

Bag 1: 4 red, 3 black. Bag 2: 2 red, 4 black. One bag selected randomly, one ball drawn. P(red)?
P = P(Bag1) x P(Red|Bag1) + P(Bag2) x P(Red|Bag2)
= 1/2 x 4/7 + 1/2 x 2/6 = 2/7 + 1/6 = 12/42 + 7/42 = **19/42**

PROBABILITY: PRACTICE QUESTIONS

PR1. Two dice thrown. P(sum = 9)?

a) 1/9
b) 1/6
c) 1/12
d) 1/36

PR2. Card drawn from 52. P(it is a face card)?

a) 1/13
b) 3/13
c) 4/13
d) 2/13

PR3. P(at least 1 head in 4 coin tosses)?

a) 15/16
b) 1/16
c) 7/8
d) 3/4

PR4. Leap year. P(53 Fridays)?

a) 1/7
b) 2/7
c) 3/7
d) 53/366

PR5. Bag has 3 red, 4 blue balls. 1 ball drawn, then 1 card from 52. P(red ball AND king)?

a) 3/91
b) 4/91
c) 1/91
d) 12/364

PR6. 6 boys, 6 girls sit in a row. P(all girls sit together)?

a) 1/66
b) 1/36
c) 1/132
d) 1/6

PROBABILITY: SOLUTIONS

PR1. Answer: a) 1/9 | Pairs: (3,6)(4,5)(5,4)(6,3) = 4. P = 4/36 = 1/9.

PR2. Answer: b) 3/13 | Face cards = J+Q+K = 12. P = 12/52 = 3/13.

PR3. Answer: a) 15/16 | P(no head) = (1/2)^4 = 1/16. P(at least 1) = 1 - 1/16 = 15/16.

PR4. Answer: b) 2/7 | Leap year: 52 weeks + 2 days. Friday in 2 of 7 possible pairs. P = 2/7.

PR5. Answer: a) 3/91 | P(red) x P(king) = 3/7 x 4/52 = 3/7 x 1/13 = 3/91. (Independent events)

PR6. Answer: c) 1/132 | Girls together: treat as 1 block. 7! x 6! arrangements. Total = 12!. P = 7!x6!/12! = 1/132.

PHASE 2: QUICK REVISION CARD

NUMBERS

1. Sum of n natural = $n(n+1)/2$ | Squares = $n(n+1)(2n+1)/6$ | Cubes = $[n(n+1)/2]^2$
2. Sum of first N odd = N^2 | Sum of first N even = $N(N+1)$
3. Factors of $a^p \times b^q \times c^r = (p+1)(q+1)(r+1)$
4. Div by 11: diff of odd-place sum and even-place sum = 0 or 11k
5. Remainder trick: multiply individual remainders, then take remainder

PERMUTATION & COMBINATION

1. AND = multiply, OR = add
2. $nPr = n!/(n-r)!$ | With repetition: $n!/p!q!r!$
3. Circular = $(n-1)!$ | Necklace = $(n-1)!/2$
4. $nCr = n!/[r!(n-r)!]$ | Select 1+ from n = $2^n - 1$
5. Handshakes = $nC2$ | Triangles = $nC3$ | 'Never together' = Total - 'Together'

PROBABILITY

1. P = Favorable/Total | P(A) + P(not A) = 1
2. P(A or B) = P(A) + P(B) - P(A and B) | If mutually exclusive: just P(A) + P(B)
3. Independent: P(A and B) = P(A) x P(B)
4. P(at least one) = 1 - P(none) <-- USE THIS ALWAYS
5. Leap year 53 of any day = 2/7 | Coins: total = 2^n | 2 Dice: total = 36

Done with Phase 2? Move to Phase 3: Work & Chain Rule, Series, Coding-Decoding